ACID REACTIONS 20

The general reactions of common acids can be summarised as follows:

acid	+	base	→	salt	+	water		
acid	+	reactive metal	→	salt	+	hydrogen		
acid	+	carbonate	→	salt	+	water	+	carbon dioxide
acid	+	hydrogencarbonate	→	salt	+	water	+	carbon dioxide

When an acid is neutralised, a salt is produced. This reaction is called a **neutralisation**. Each acid produces a certain type of salt. The salt produced contains most of the elements that were present in the acid from which it was made. Salts do not contain hydrogen ions – these have been replaced by metal ions. Common table salt is called *sodium chloride* and has the formula NaCl.

Hydrochloric acid (HCl) forms salts called chlorides ($-Cl^-$).

Nitric acid (HNO_3) forms salts called nitrates ($-NO_3^-$).

Sulfuric acid (H_2SO_4) forms salts called sulfates ($-SO_4^{2-}$).

Phosphoric acid (H_3PO_4) forms salts called phosphates ($-PO_4^{3-}$).

Acid + base

You can tell when an acid has been neutralised by a base (or vice versa) by adding an indicator. An indicator is a dye that changes colour to indicate whether a substance is acidic or alkaline (basic).

1 Write a word equation and then a balanced equation for each reaction between acid and base.

- a hydrochloric acid and sodium hydroxide (NaOH)
- b sulfuric acid and copper II oxide (CuO)
- c nitric acid and ammonium hydroxide (NH_4OH)
- d phosphoric acid and potassium hydroxide (KOH)

Acid + metal

When reactive metals are added to acids, hydrogen gas (H_2) and a salt are formed. Hydrogen is an invisible, very flammable, lighter-than-air gas; it goes 'POP!' when placed next to a flame. Unreactive metals such as gold (Au) tend not to react with acids.

2 Write a word equation and then a balanced equation for each reaction between acid and metal.

- a hydrochloric acid and magnesium
- b sulfuric acid and aluminium
- c nitric acid and zinc
- d phosphoric acid and iron (assume iron forms the Fe^{3+} ion)

Acid + carbonate

When acids and carbonates react, carbon dioxide (CO_2) gas is given off and water and a salt are formed. Hydrogencarbonates react in a similar manner. When carbon dioxide gas is bubbled through limewater, the limewater changes from clear to cloudy/milky.

3 Write a word equation and then a balanced equation for each reaction between acid and carbonate or hydrogencarbonate.

- a hydrochloric acid and copper (II) carbonate ($CuCO_3$)
- b sulfuric acid and potassium carbonate (K_2CO_3)
- c nitric acid and sodium hydrogencarbonate ($NaHCO_3$)
- d phosphoric acid and potassium hydrogencarbonate ($KHCO_3$)

4 Describe a chemical test for the presence of carbon dioxide.

5 'A compound that is formed by the replacement of the hydrogen atom(s) of an acid molecule with a metal or a group acting like a metal.' What does this statement define?

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